

Comparison of Static Characteristics in GaN HEMTs Across 50K to 400K Considering Diverse Techniques and Statistical Variation

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Abstract—GaN HEMTs are prominent candidates for space applications, such as lunar missions, which require reliable operation over a wide temperature range from 50K to 400K. Previous studies have examined the performance of GaN HEMTs at cryogenic temperatures (typically down to 77K), but they often focus on a limited selection of models and sample sizes. To address this gap, a comprehensive analysis of diverse GaN HEMTs is necessary, considering their performance across a broad temperature range and statistical variations due to manufacturing inconsistencies. This paper provides a comparison of GaN HEMTs based on different voltage ratings, technologies and packaging approaches. A key observation is that the on-state resistance of the studied Gate Injection Transistor (GIT) device exhibits a parabolic variation with temperature, with resistance at 50K being 1.25 times higher than at room temperature. Additionally, the statistical variation in the static parameters of a cascode GaN device across six samples reveals that the variation increases as the temperature decreases, indicating that the statistical variation over a wide temperature range cannot be overlooked.

Index Terms—static characterization, GaN HEMT, GIT (Gate Injection Transistor), D-mode GaN, EPC, GaN Systems, technology, on-state resistance, cryogenic temperature, leakage current

I. INTRODUCTION

Wide-bandgap devices, such as SiC and GaN, are among the most promising alternatives to traditional Si MOSFETs and IGBTs. These devices are increasingly being adopted to replace conventional Si-based converters due to their ability to enhance efficiency and support high power density. This superior performance stems from several key advantages: a high critical electric field, faster switching speeds, superior

thermal conductivity, lower on-state resistance, and reduced power losses [1].

Among these technologies, GaN HEMTs exhibit enhanced mobility and low on-state resistance, particularly at low temperatures. Unlike Si and SiC, GaN HEMTs avoid carrier freeze-out at cryogenic temperatures because they rely on a two-dimensional electron gas (2DEG) formed at the Al-GaN/GaN heterojunction. This 2DEG is induced by spontaneous and piezoelectric polarization fields rather than by inversion, enabling GaN HEMTs to achieve high conductivity in cryogenic environments. In contrast, silicon and SiC devices suffer from carrier freeze-out at low temperatures due to insufficient thermal energy to ionize dopant atoms, which results in reduced conductivity. Additionally, SiC devices are further impacted by interface traps, which increase on-state resistance and threshold voltage under cryogenic conditions [2]–[4]. Silicon MOSFETs, on the other hand, experience a reduction in breakdown voltage at low temperatures. This is because the mean free path of carriers increases, allowing them to gain more energy for a given electric field before collisions, thus resulting in a reduced avalanche breakdown voltage [5].

Furthermore, GaN HEMTs are being considered for space applications that require reliable operation over a broad temperature range, such as lunar missions from 50K to 400K [6]. While extensive characterization of GaN HEMTs under cryogenic conditions has been conducted in previous studies, these studies often involve small sample sizes and temperature measurements that typically extend only down to 77K [7]–[9]. Some research attempts to compare GaN devices based on gate technology; however, such comparisons are limited by small sample sizes, lack of characterization at 50K, and

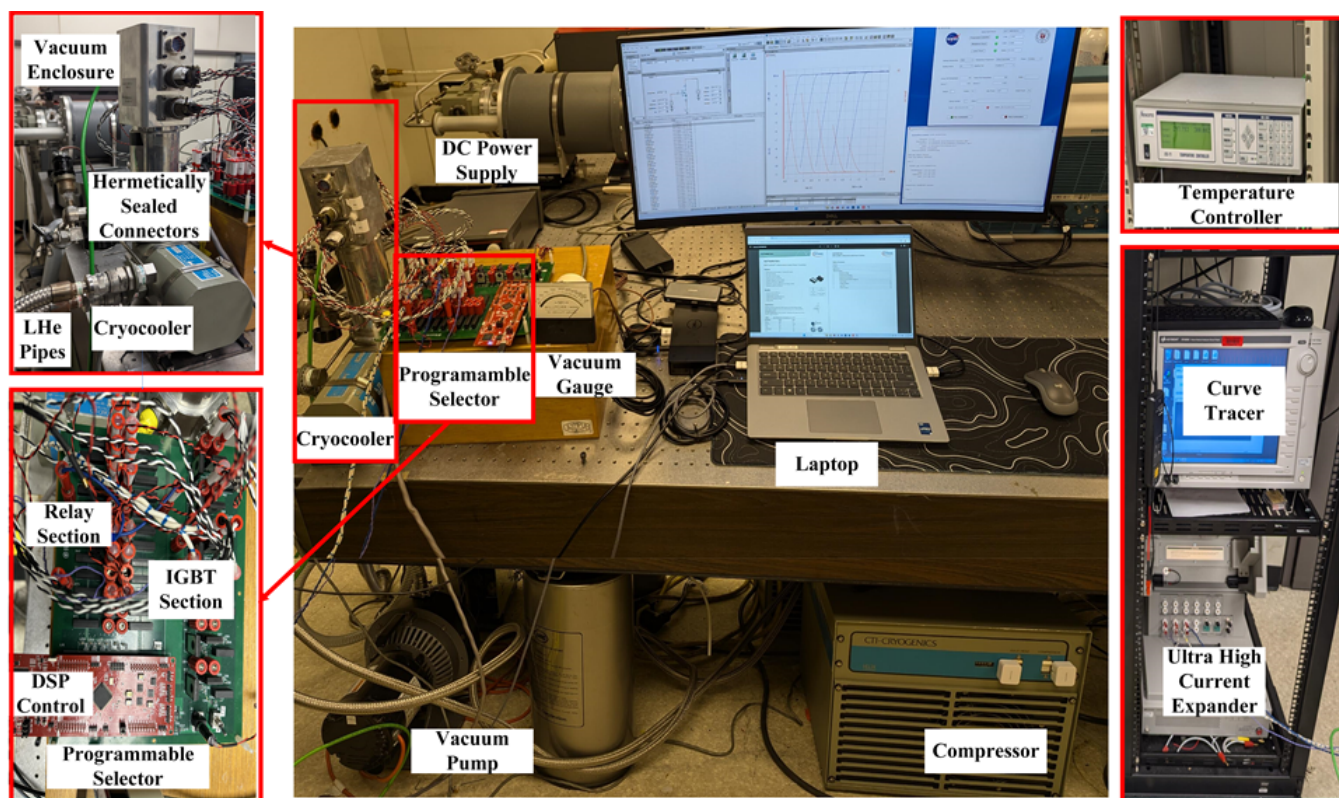


Fig. 1. Hardware overview of static characterization platform.

insufficient consideration of statistical variations [10]–[13]. For instance, Nela investigated the static characteristics of GaN devices down to 4.2K, but this study was restricted to only two samples and focused exclusively on devices in the 600V–650V range [14]. Many papers have attempted to model the behavior of GaN devices for a particular technology, focusing either on high-temperature variations or limited temperature ranges. However, these studies often focus on specific technologies, which do not comprehensively represent GaN device performance under cryogenic conditions.

The limitations of current state-of-the-art studies highlight the need for a comprehensive comparative analysis of GaN HEMTs across a wide temperature range, considering the often-overlooked impact of statistical parameter variations among samples. This paper addresses these gaps by systematically examining the temperature-dependent behaviors (from 50K to 400K) of static parameters in a diverse array of GaN HEMTs, with holistic consideration of GaN semiconductor chips, packaging technologies, voltage ratings, and statistical variations.

First, the hardware setup for the static characterization platform is overviewed. Then, the impacts of different chip, packaging technologies and voltage ratings dependence on temperature are discussed, respectively. Next, the statistical variation among different samples of the same device is analyzed. Finally, key takeaways and future work are summarized.

II. AUTOMATED STATIC CHARACTERIZATION PLATFORM

A fully automated platform, consisting of a cryocooler, curve tracer, programmable selector, temperature controller, and a computer (as shown in Fig. 1), has been developed to characterize the static behaviors of GaN HEMTs across a temperature range from 50K to 400K. The detailed setup description of the static characterization platform can be found in [15], and the lead optimization inside the cryocooler for this setup can be found in [16]. Multiple devices can be characterized simultaneously on this platform. The devices under test are placed inside the cryocooler, and the temperature is varied from 50K to 400K using the temperature controller. The vacuum inside the cryocooler is maintained by a vacuum pump, and the pressure is monitored using a vacuum gauge.

The devices are characterized using the curve tracer. To switch between devices for characterization, the relay board employs relays and IGBT switches. The relays are used for low-current signals, such as gate and sense wires, while the IGBTs handle high-current force wires. These components are controlled by a microcontroller. The entire automation process—device selection, temperature adjustment, static characterization initiation, data saving, and data processing—is managed using a MATLAB GUI.

Table I summarizes all the GaN devices from various vendors that were characterized across the 50K to 400K temperature range with complete data.

III. DEPENDENCE ON VOLTAGE RATINGS

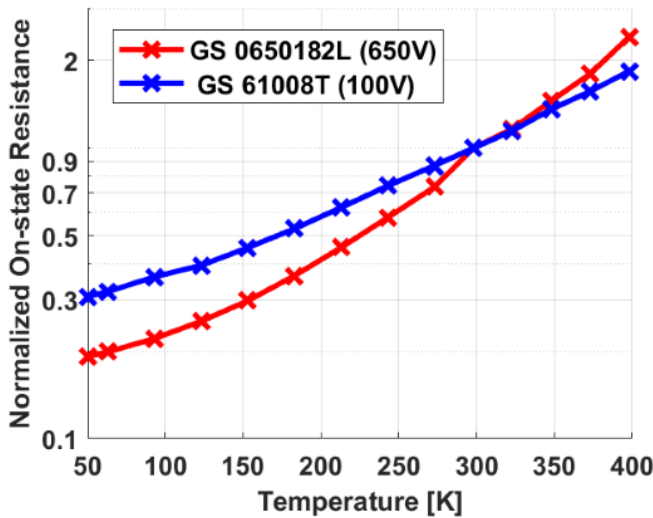


Fig. 2. Normalized on state resistance of GS0650182L [650V] and GS61008T [100V].

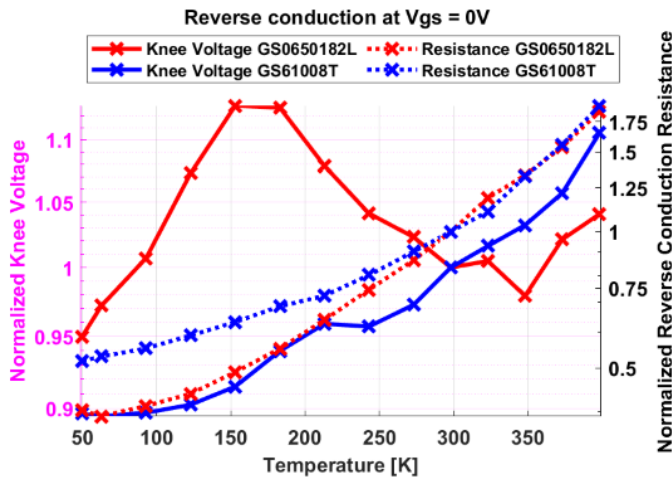


Fig. 3. Normalized knee voltage and reverse conduction resistance of GS0650182L [650V] and GS61008T [100V].

In this paper, we have used normalized values of static parameters such as on-state resistance, knee voltage, and leakage current. The normalized value of a static parameter is defined as the ratio of the parameter at a given temperature to its corresponding value at room temperature.

Fig. 2, Fig. 3, and Fig. 4 compare the variation in static parameters between the GS0650182L and GS61008T devices, both of which utilize the same enhancement-mode GaN HEMT chip technology from the same manufacturer. The GS0650182L is rated for 650V, while the GS61008T is rated for 100V.

Fig. 2 shows that both devices exhibit a decrease in normalized on-state resistance as the temperature decreases. This

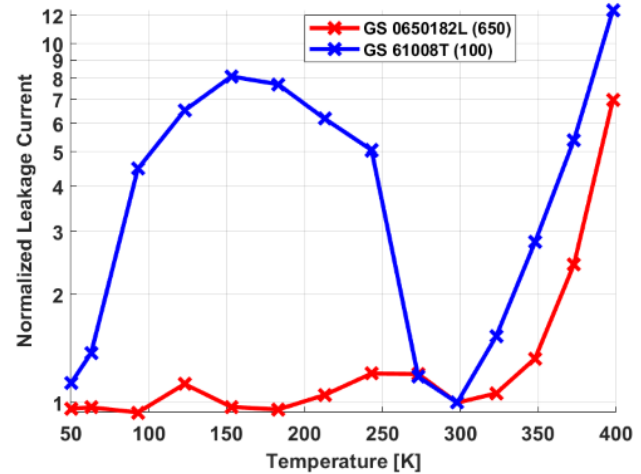


Fig. 4. Normalized leakage current of GS0650182L [650V] and GS61008T [100V].

trend is primarily attributed to two factors: the reduction in resistance of the 2DEG regions due to improved electron mobility at lower temperatures, and the decrease in metal contact and package resistance with lower temperatures.

The GS0650182L (650V) exhibits a steeper reduction in normalized on-state resistance compared to the GS61008T (100V). At 50K, the normalized on-state resistance of the 650V device is 0.19, significantly lower than that of the 100V device (0.31). This observation suggests that higher voltage-rated devices are more sensitive to temperature variations.

The normalized knee voltage of the GS61008T during reverse conduction consistently decreases with temperature. In contrast, the variation in the normalized knee voltage for the GS0650182L is inconsistent, as shown in Fig. 3.

Fig. 4 shows that the leakage current for the 650V device remains stable at lower temperatures, whereas the leakage current for the 100V device varies significantly. This behavior indicates that the leakage current for the 100V device is more sensitive to temperature variations and suggests greater susceptibility of its breakdown voltage (BV) to temperature changes.

IV. DEPENDENCE ON CHIP TECHNOLOGIES

Fig. 5, Fig. 6, and Fig. 7 illustrate the variation in static parameters for the IGOT60R070D1, GS0650182L, and TP65H070G4LSG devices, all of which belong to the 600–650 V family, originate from different vendors, and utilize distinct chip technologies.

The p-GaN HEMT (GS0650182L) was discussed in the previous section, where it was observed that the on-state resistance decreases with temperature. Additionally, the knee voltage during reverse conduction exhibits inconsistent behavior, while the leakage current shows minimal variation at lower temperatures.

For the GIT device (IGOT60R070D1), the variation in normalized on-state resistance with temperature follows a

TABLE I
COMMERCIALLY AVAILABLE GAN HEMTs: THEIR VOLTAGE AND RESISTANCE RATINGS AT ROOM TEMPERATURE (25°C), TECHNOLOGY TYPE, AND PACKAGING.

Sr. No	Vendor	Part Number	Ratings (@25 °C) Voltage / Resistance	Technology	Package
1	GaN Systems	GS61008T	100V / 7mΩ	E-mode GaN HEMT (p-GaN)	GaNPx packaging
2	EPC	EPC2215	200V / 6mΩ	E-mode GaN HEMT (p-GaN)	LGA
3	EPC	EPC2034C	200V / 8mΩ	E-mode GaN HEMT (p-GaN)	BGA
4	EPC	EPC2307	200V / 8.2mΩ	E-mode GaN HEMT (p-GaN)	QFN
5	Infineon	IGOT60R070D1	600V / 55mΩ	GIT, E-mode GaN HEMT (p-GaN gate with Ohmic Gate)	PG-DSO
6	Transphorm	TP65H070LSG	650V / 72mΩ	D-mode GaN HEMT w/ Si for normally-off cascode	PQFN
7	GaN Systems	GS-065-018-2-L	650V / 78mΩ	E-mode GaN HEMT (p-GaN)	PDFN packaging

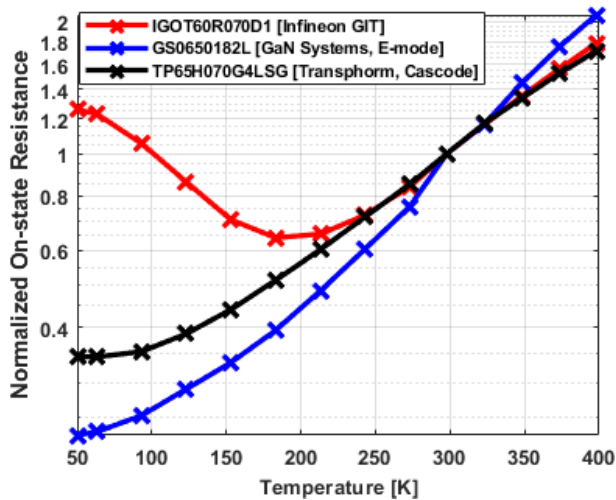


Fig. 5. Normalized on state resistance of IGOT60R070D1 [p-GaN with Ohmic gate], GS0650182L [p-GaN], TP65H070G4LSG [Cascode].

parabolic trend. The resistance decreases as the temperature drops below room temperature, reaching a minimum value at around 183K, and then begins to increase. This behavior can be attributed to the carrier freeze-out of dopants in the p-GaN layer. At 50K, the normalized on-state resistance is approximately 1.25 times higher than at room temperature, indicating reduced efficiency for power converter applications at low temperatures. Despite this, the drain leakage current remains stable under cryogenic conditions. Additionally, the knee voltage for this device increases as the temperature decreases.

For the cascode device (TP65H070G4LSG), the on-state

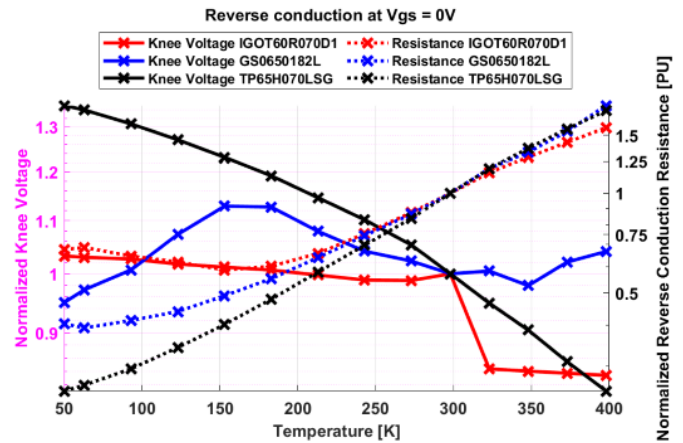


Fig. 6. Normalized knee voltage and reverse conduction resistance of IGOT60R070D1 [p-GaN with Ohmic gate], GS0650182L [p-GaN], TP65H070G4LSG [Cascode].

resistance decreases with temperature, consistent with the expected behavior of Si and GaN devices. However, compared to GIT devices, the decrease is less steep. Below 100K, the slope of the resistance-temperature curve becomes nearly constant, suggesting limited improvement in conductivity due to carrier freeze-out in the silicon MOSFET component of the device. Furthermore, the leakage current for the cascode device increases at cryogenic temperatures, aligning with the behavior observed in silicon devices, where leakage current typically rises as temperature decreases.

V. DEPENDENCE ON PACKAGING TECHNOLOGIES

Fig. 8, Fig. 9, and Fig. 10 illustrate the variation in static parameters for the EPC 2215, EPC 2034C, and EPC 2307

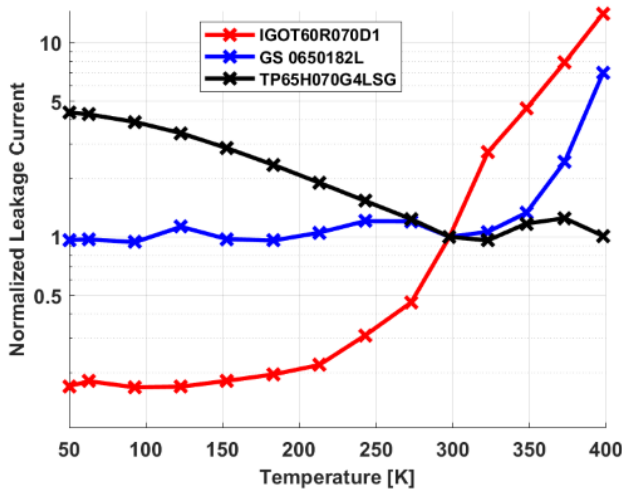


Fig. 7. Normalized leakage current of IGOT60R070D1 [p-GaN with Ohmic gate], GS0650182L [p-GaN], TP65H070G4LSG [Cascode].

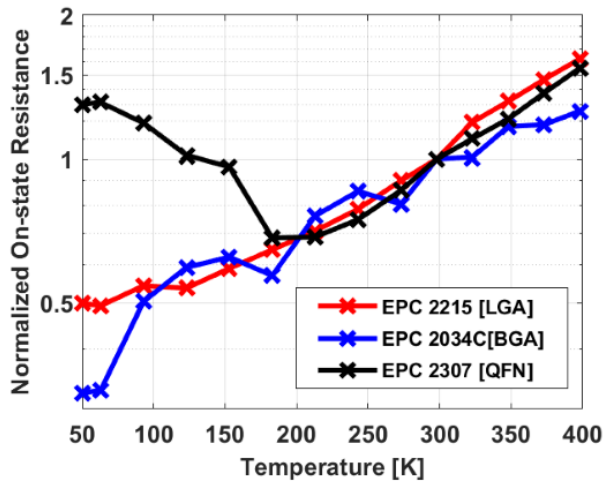


Fig. 8. Normalized on-state resistance of EPC 2215 [LGA], EPC 2034C [BGA], EPC 2307 [QFN].

devices. All three devices have the same voltage rating of 200V and are manufactured by the same vendor (EPC) but differ in packaging: LGA, BGA, and QFN, respectively.

The normalized on-state resistance for the EPC 2215 and EPC 2034C follows a consistent decreasing trend with temperature, as expected for p-GaN devices. However, the EPC 2307, which uses a QFN package, exhibits an anomalous trend. Its normalized on-state resistance decreases with temperature down to 183K, where it reaches a minimum, and then begins to increase. At 50K, the on-state resistance for the EPC 2307 is approximately 1.31 times higher than at room temperature, which is atypical for p-GaN-based devices.

The variation in reverse conduction resistance across the three devices consistently shows a decreasing trend with temperature. The reverse knee voltage demonstrates little sensitivity at temperatures near room temperature but becomes

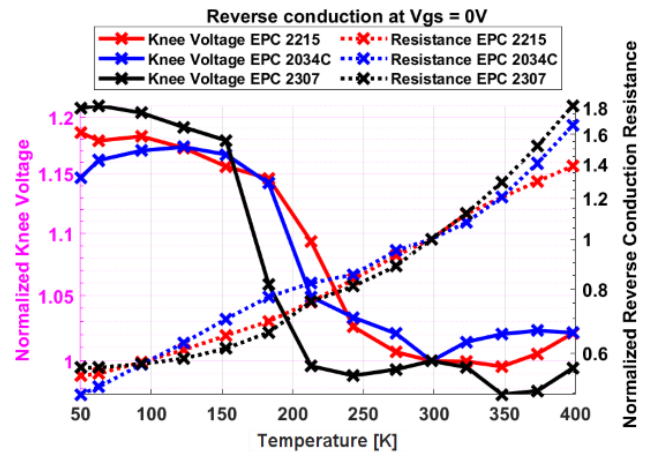


Fig. 9. Normalized knee voltage and reverse conduction resistance of EPC 2215 [LGA], EPC 2034C [BGA], EPC 2307 [QFN].

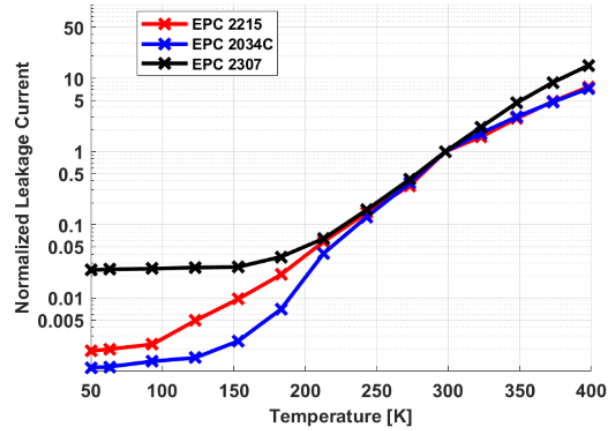


Fig. 10. Normalized leakage current of EPC 2215 [LGA], EPC 2034C [BGA], EPC 2307 [QFN].

increasingly sensitive as the temperature drops further.

The parabolic nature of the $R_{DS(on)}$ trend for the EPC 2307 is unusual for EPC devices. Whether this behavior is due to the packaging or other factors requires further investigation.

VI. QUANTIFICATION OF STATISTICAL VARIATION

Fig. 11, Fig. 12, and Fig. 13 illustrate the statistical variation in the static parameters of the TP65H070G4LSG (cascode device) across six samples. The data indicate that the variation in static parameters, such as $R_{DS(on)}$, knee voltage, and reverse conduction resistance, is greater at 50K compared to room temperature. Additionally, the variation in leakage current becomes most pronounced at 400K. At 50K, the tolerances for the static parameters are as follows: on-state resistance ($R_{DS(on)}$) ± 1.11 m Ω , knee voltage ± 0.02 V, and leakage current ± 1.98 μ A, relative to their respective mean values of 18.64 m Ω , 1.23 V, and 9.59 μ A.

The results indicate that the statistical variation among samples from the same device cannot be overlooked, as it

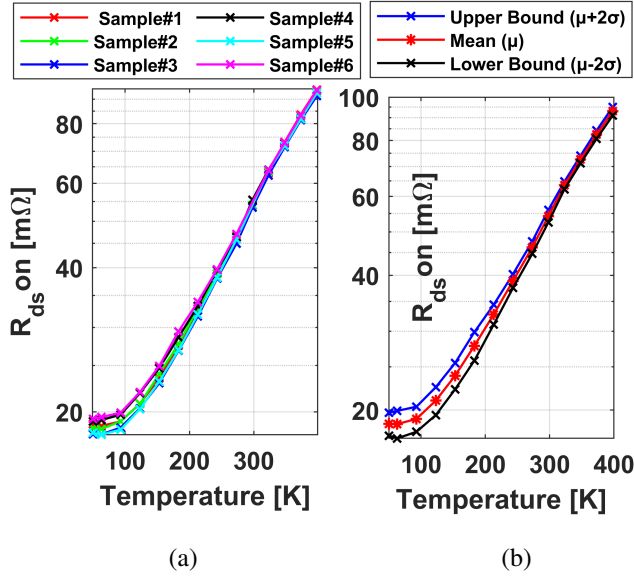


Fig. 11. On state resistance of TP65H070G4LSG. (a) Individual values of the 6 devices. (b) Calculated confidence interval.

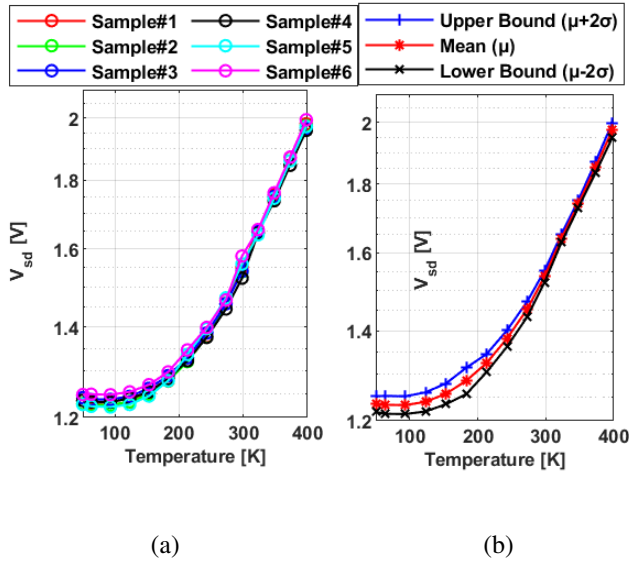


Fig. 12. Knee voltage of TP65H070G4LSG. (a) Individual values of the 6 devices. (b) Calculated confidence interval.

is particularly important for high-power modules integrating multiple chips. Deviations in static parameters can lead to imbalances in current sharing, potentially causing premature failure. Ensuring uniformity in these parameters is essential for robust, efficient, and reliable circuit performance.

VII. CONCLUSION

In this paper, we analyzed the static performance of various commercially available GaN HEMTs over a wide temperature range, from 50K to 400K. The comparison included devices with different voltage ratings, chip technologies, and pack-

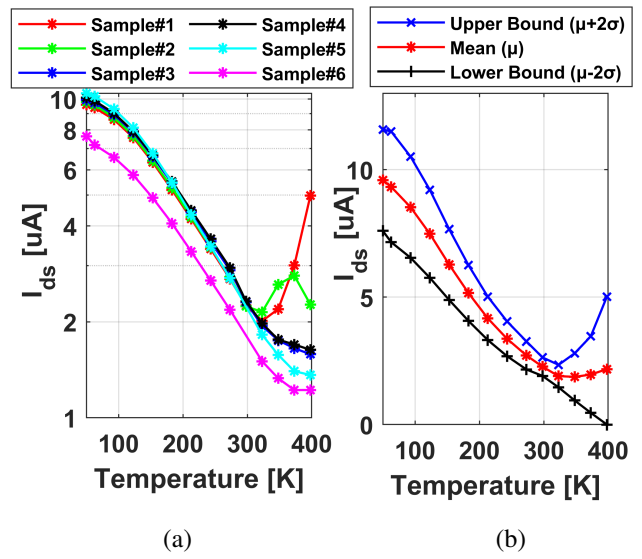


Fig. 13. Leakage current of TP65H070G4LSG. (a) Individual values of the 6 devices. (b) Calculated confidence interval.

aging approaches. At 50K, the 600V GaN device from GaN Systems exhibited greater temperature sensitivity in $R_{DS(on)}$ compared to the 100V device from the same manufacturer. Despite this, the 600V device demonstrated a lower normalized on-state resistance and more consistent leakage current behavior at cryogenic temperatures.

For 650V GaN devices employing different technologies (p-GaN HEMT, p-GaN with Ohmic gate (GIT), and cascode configurations), we observed notable distinctions in temperature-dependent performance. The on-state resistance of the p-GaN HEMT decreased consistently as the temperature dropped. In contrast, GIT device exhibited a parabolic trend, with its $R_{DS(on)}$ at 50K being approximately 1.25 times higher than at room temperature. This indicates higher conduction losses and reduced efficiency under cryogenic conditions, making it less suitable for power conversion at low temperatures.

For devices from the same 200V family manufactured by EPC, we investigated the impact of packaging (QFN, BGA, and LGA). Significant performance differences were observed, with the EPC2307 (QFN package) exhibiting an unusual $R_{DS(on)}$ trend compared to other EPC devices. Whether this behavior is due to packaging or other factors requires further investigation.

Finally, we found that the statistical variation in device performance at 50K was more pronounced than at room temperature. This increased variability is critical for applications involving parallel operation of GaN HEMTs under cryogenic conditions, as it directly affects reliability and uniformity of performance.

In the future, we plan to further investigate the unusual behavior of the EPC2307 and expand our analysis by characterizing additional samples to better understand statistical variations. Additionally, we aim to perform dynamic charac-

terization and develop a comprehensive model for GaN devices that accounts for wide temperature ranges (50K to 400K) and statistical variations.

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